

Measurement of Absolute Double Differential Cross-Section in Invariant Mass and p_T Bins

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Input Files and Bin Ranges

Input Data Files:

- Merged ROOT files containing reconstruction and hodoscope efficiencies.
- **Roadsets:** RS57, RS59, RS62, RS67 (run 3089), and RS70.
- **Targets:** Liquid Hydrogen (LH2), Liquid Deuterium (LD2), and Empty Flask.

Kinematic Binning Definition:

- **Mass Bins (GeV):** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.7]
- **p_T Bins (GeV):** [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]
- **x_F Bins:** [0.0, 0.85] with a fixed step width of 0.05.

Event Selection Criteria

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8 \text{ GeV}$, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, and $|\cos \theta| < 0.5$.
- Vertex cuts: $-280 < dz < -5$, target/dump tracking separation criteria met.
- Transverse momentum bounds: $|dp_x| < 1.8$, $|dp_y| < 2.0$, and $dp_x^2 + dp_y^2 < 5$.

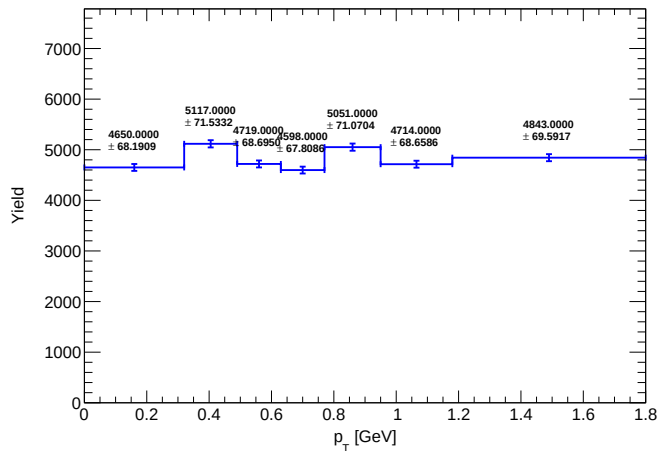
Track Quality & Acceptance:

- $\chi^2_{\text{target}} < 15$, $\chi^2_{\text{dimuon}} < 18$, and $\chi^2/(\text{nHits} - 5) < 12$.
- Hit requirements: $\text{nHits} > 13$ per track, $\text{nHits}_1 + \text{nHits}_2 > 29$, $\text{St1 total} > 8$.
- Track vertex longitudinal limits: $-320 < z_v < -5$.
- A dynamic beam offset correction is applied to y-coordinates: offset is 1.6 for $\text{runID} \geq 11000$ and 0.4 otherwise.

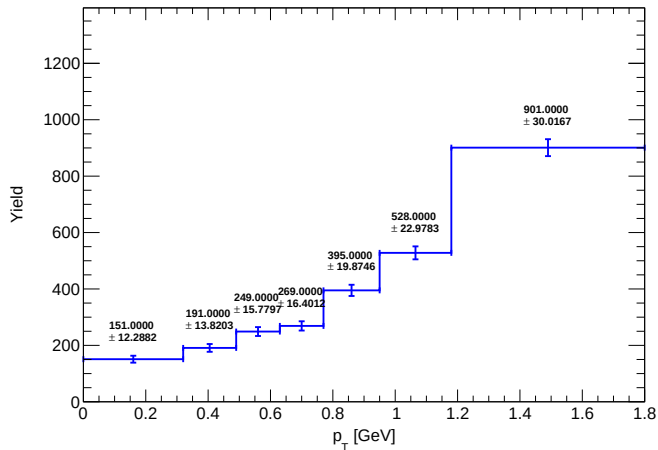
Detector Occupancy Limits:

- Drift chamber hits: $D1 < 400$, $D2 < 400$, $D3 < 400$, and total $D1 + D2 + D3 < 1000$.

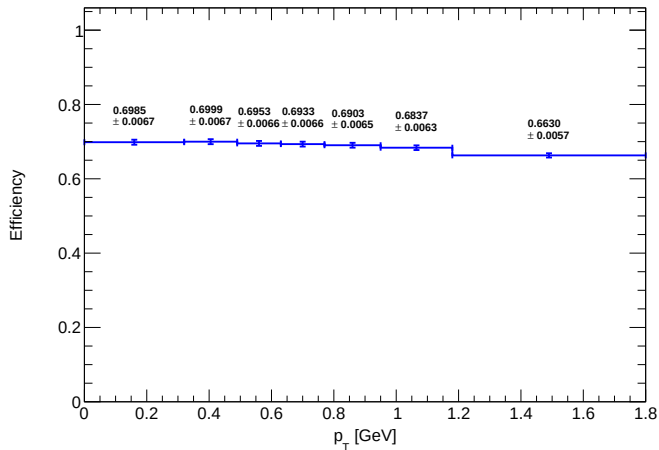
Total Yield (result) (LH2)



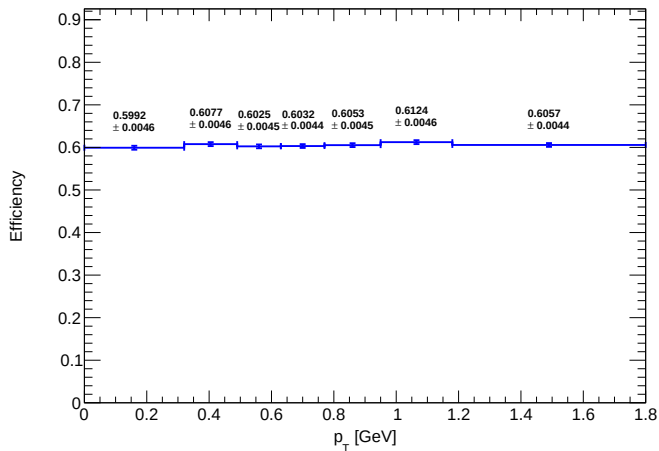
Mix Yield (result_mix) (LH2)



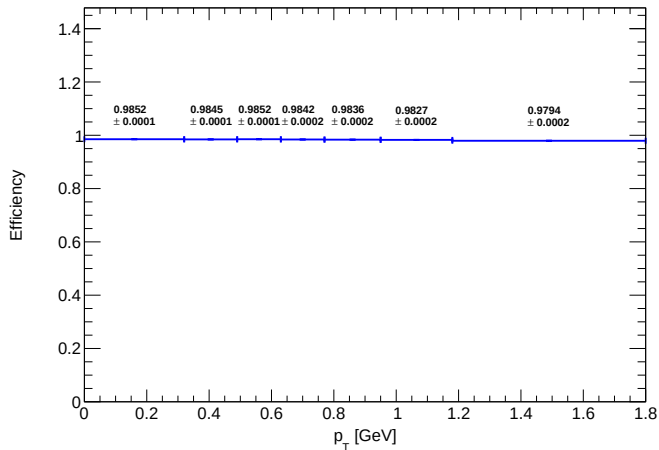
Avg Reco Eff (Total) (LH2)



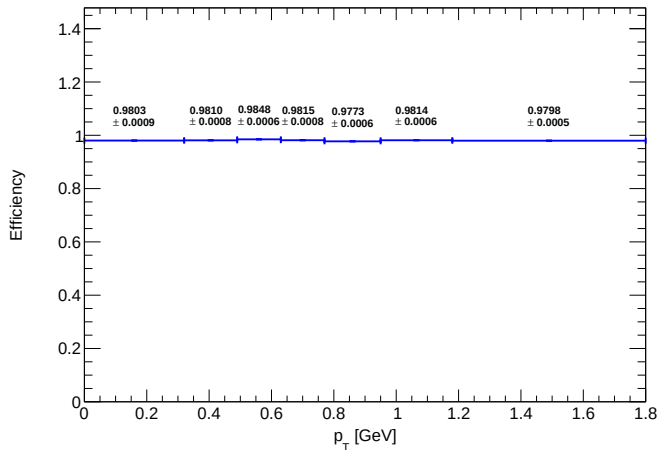
Avg Reco Eff (Mix) (LH2)



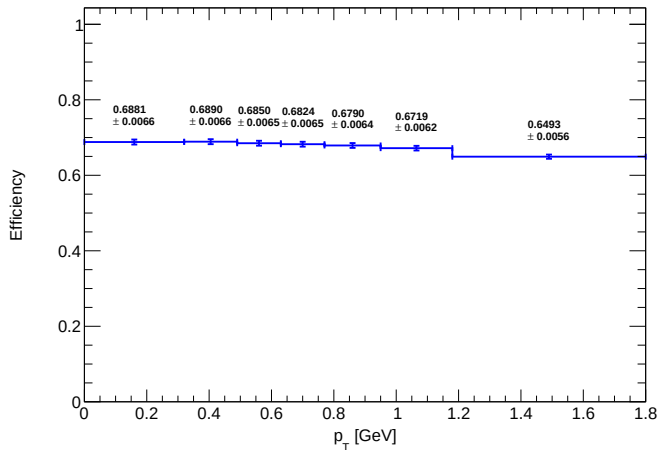
Avg Hodo Eff (Total) (LH2)

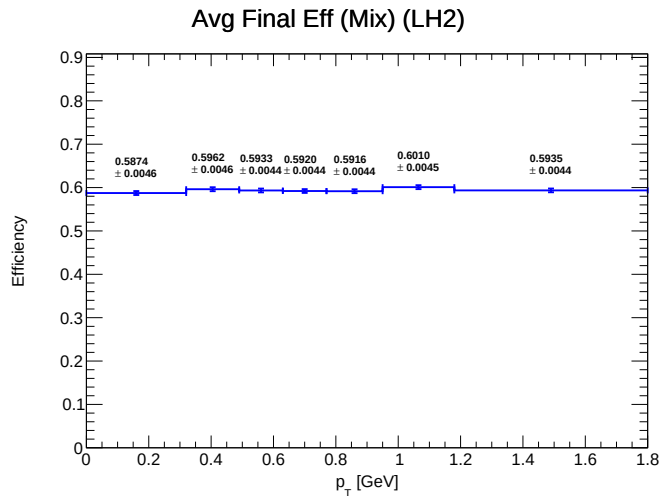


Avg Hodo Eff (Mix) (LH2)

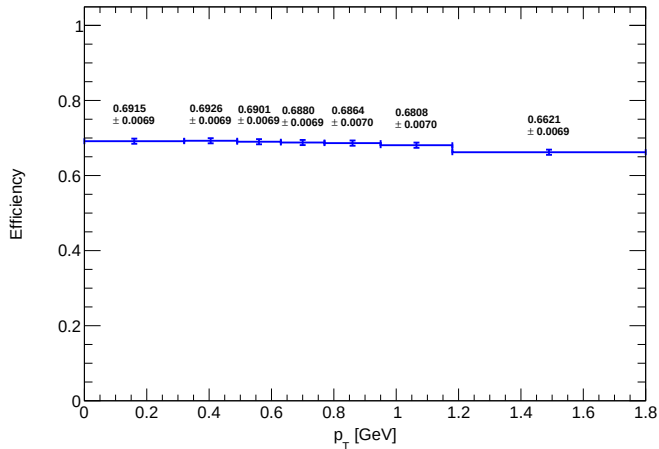


Avg Final Eff (Total) (LH2)

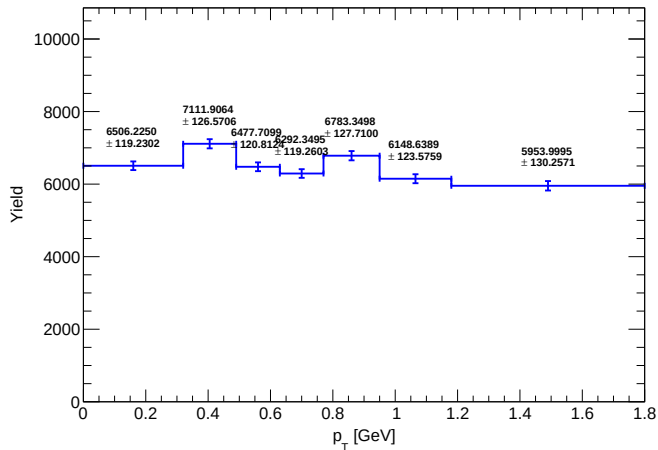




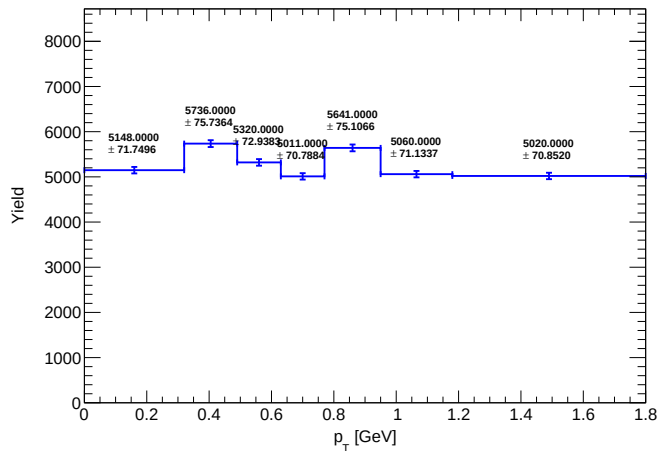
Avg Signal Efficiency (LH2)



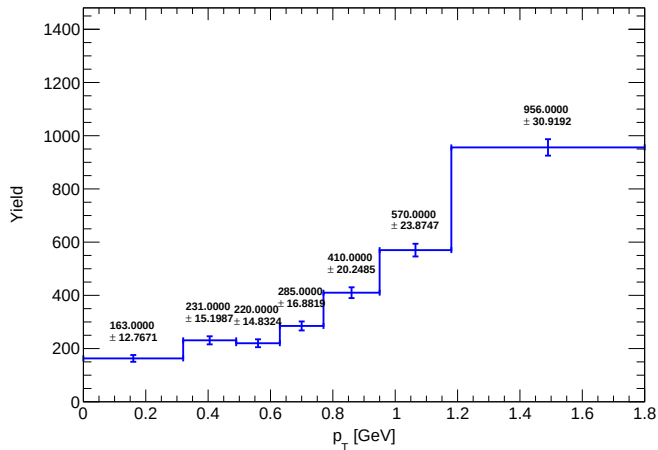
Corrected Yield (Total Error) (LH2)



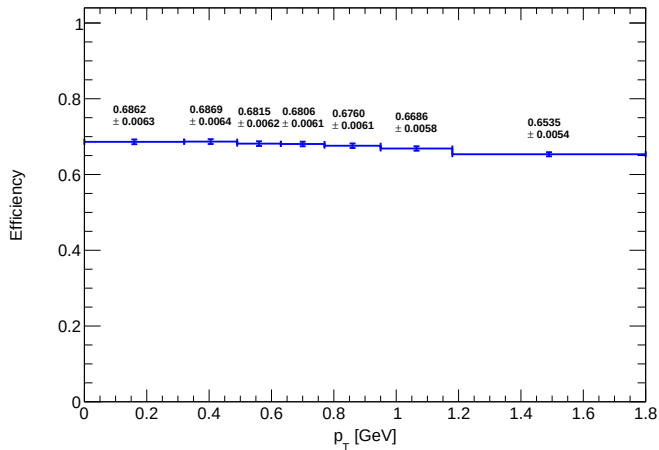
Total Yield (result) (LD2)



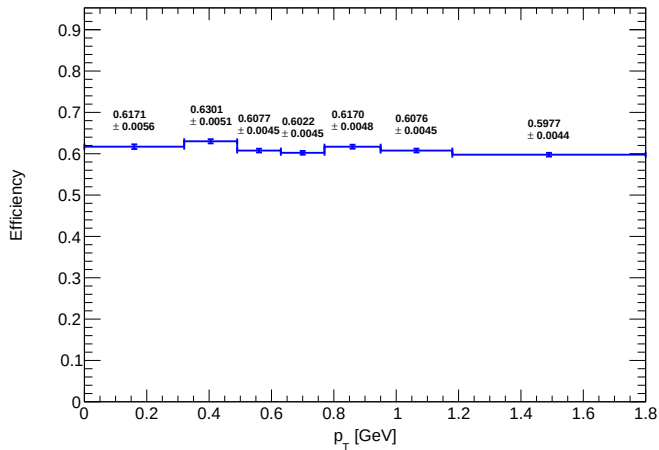
Mix Yield (result_mix) (LD2)



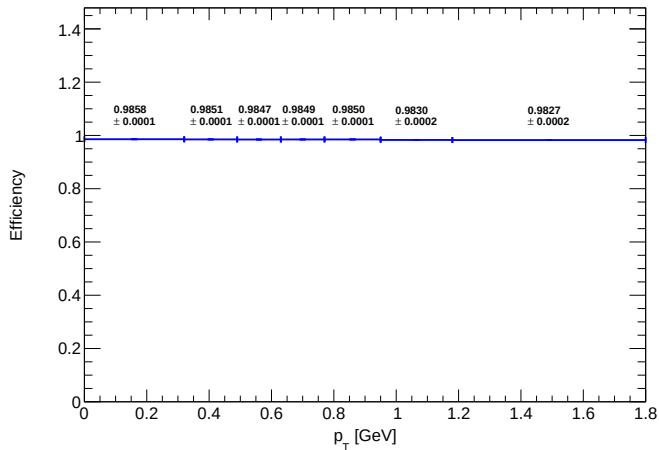
Avg Reco Eff (Total) (LD2)



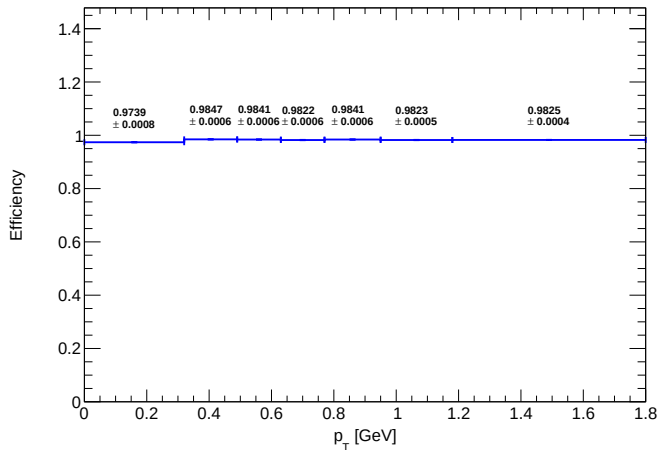
Avg Reco Eff (Mix) (LD2)



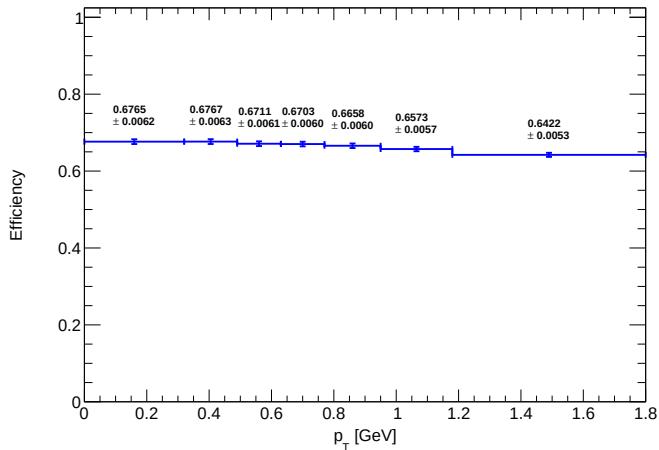
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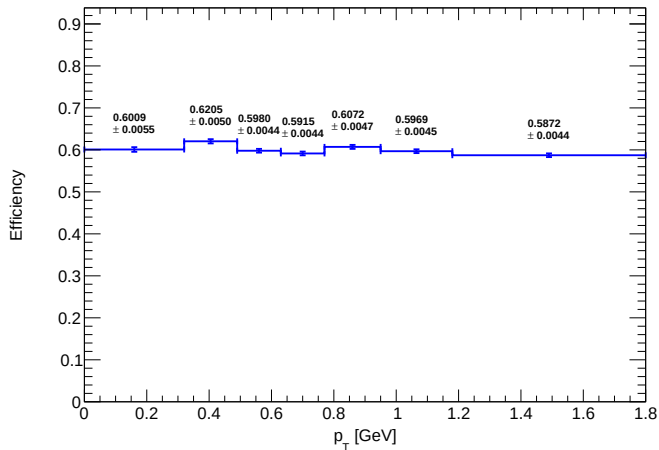
Avg Hodo Eff (Mix) (LD2)



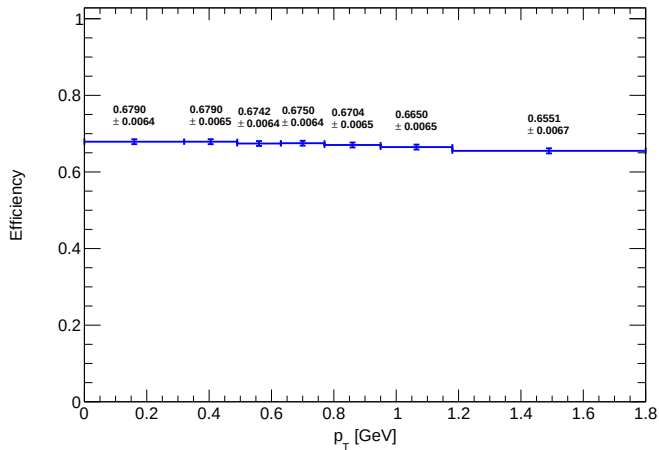
Avg Final Eff (Total) (LD2)



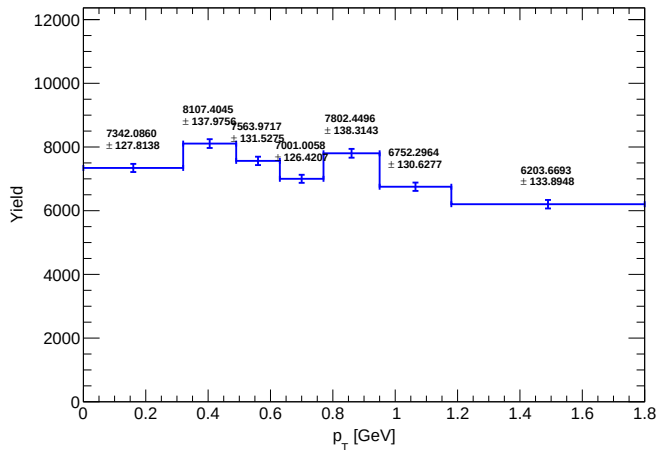
Avg Final Eff (Mix) (LD2)



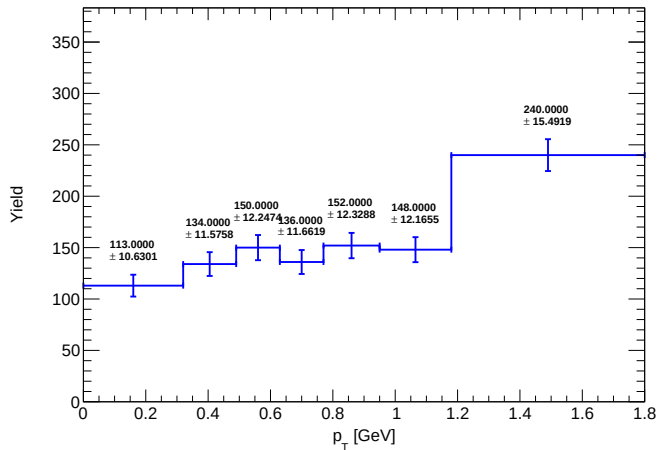
Avg Signal Efficiency (LD2)



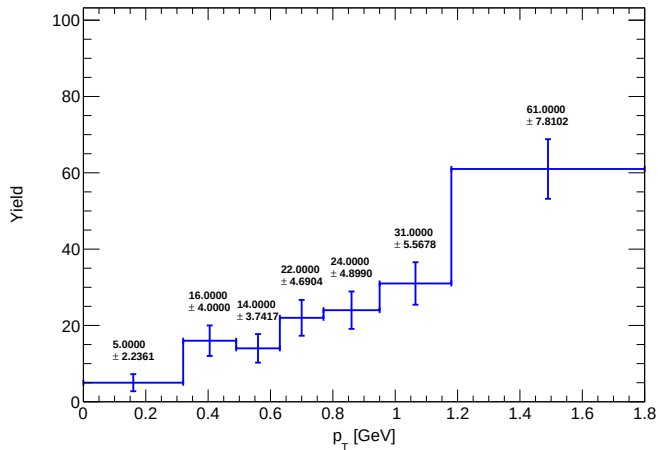
Corrected Yield (Total Error) (LD2)



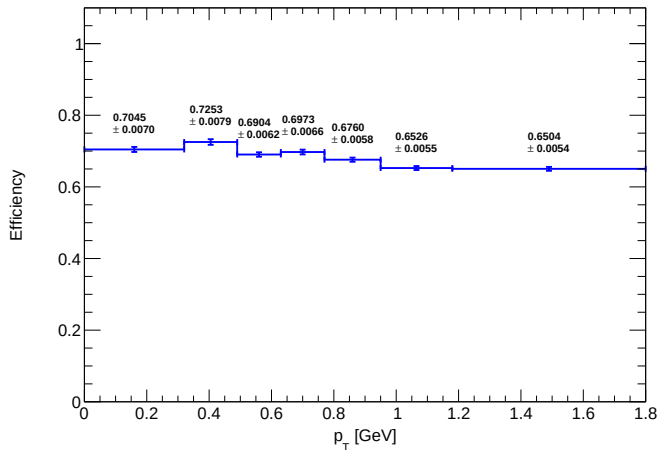
Total Yield (result) (Flask)



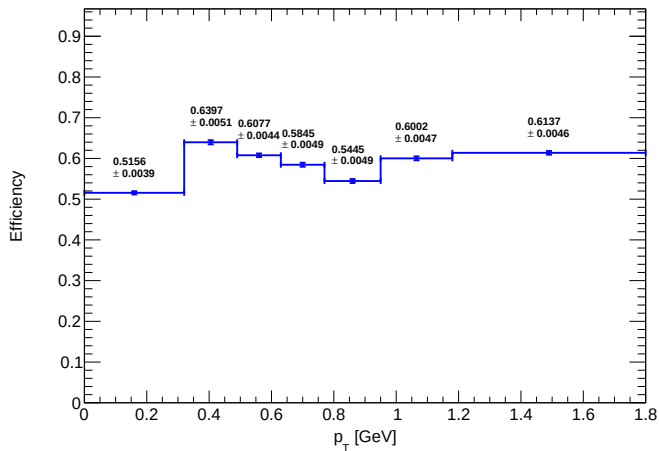
Mix Yield (result_mix) (Flask)



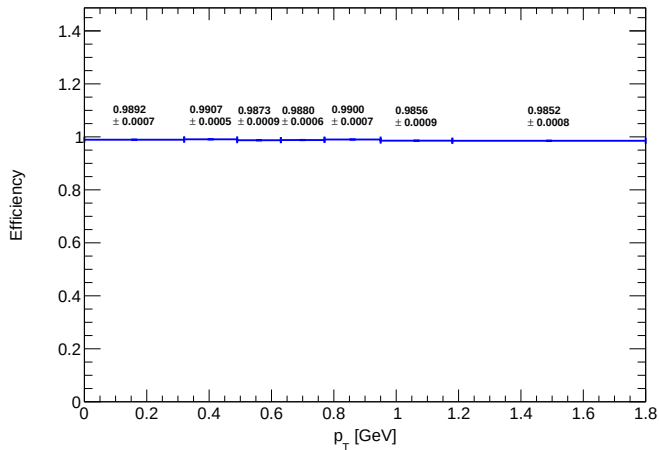
Avg Reco Eff (Total) (Flask)



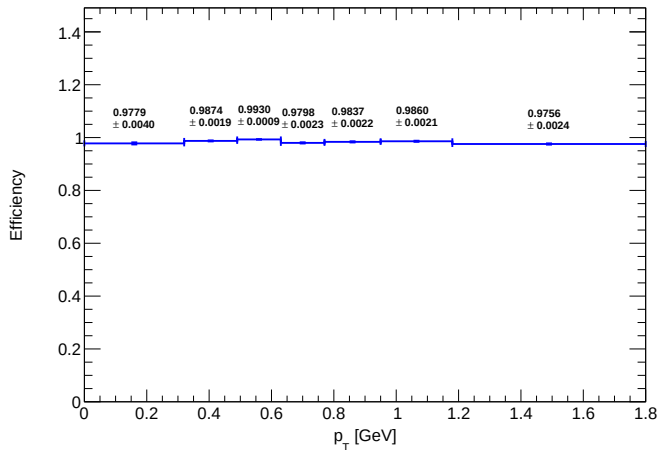
Avg Reco Eff (Mix) (Flask)



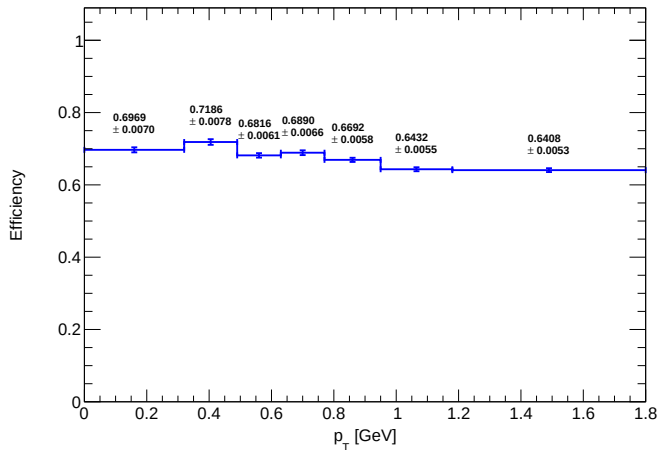
Avg Hodo Eff (Total) (Flask)



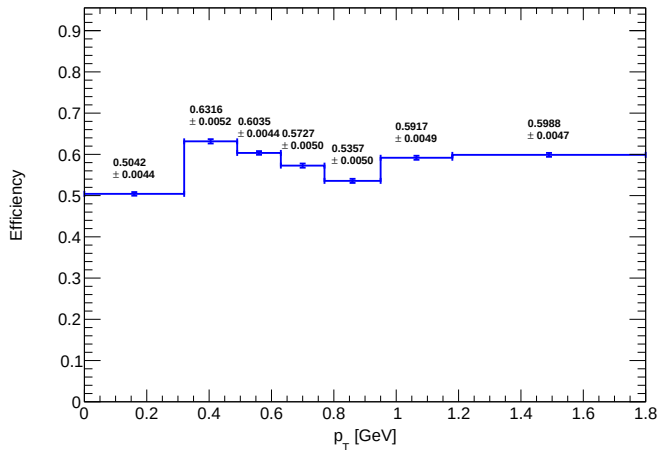
Avg Hodo Eff (Mix) (Flask)



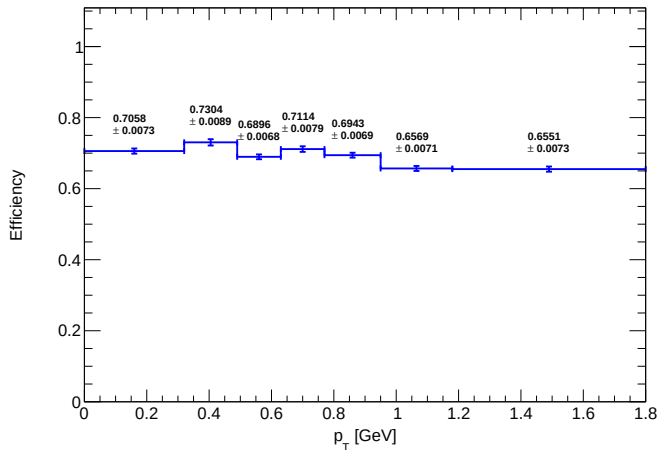
Avg Final Eff (Total) (Flask)



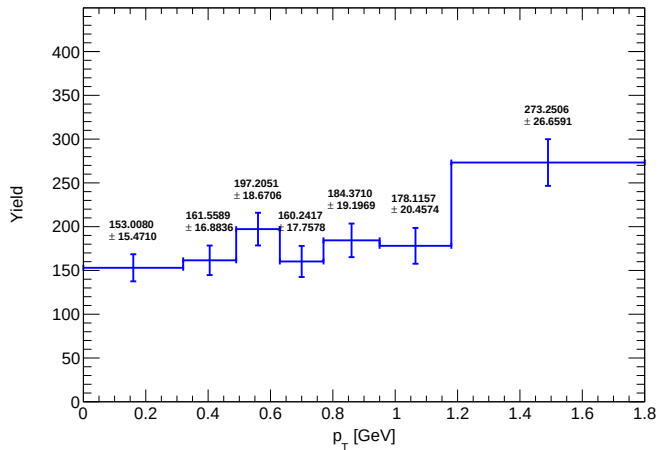
Avg Final Eff (Mix) (Flask)



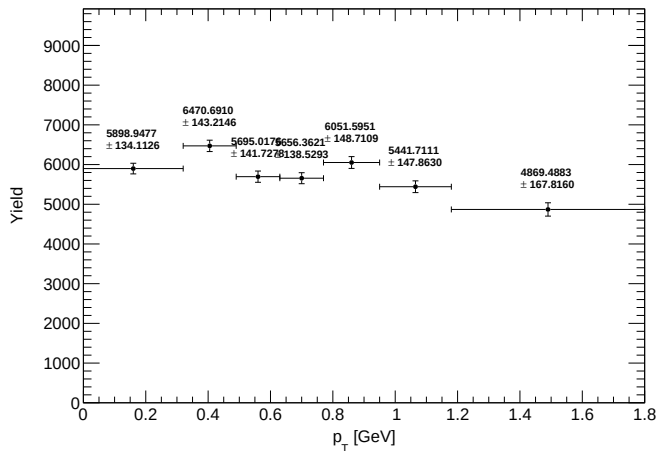
Avg Signal Efficiency (Flask)



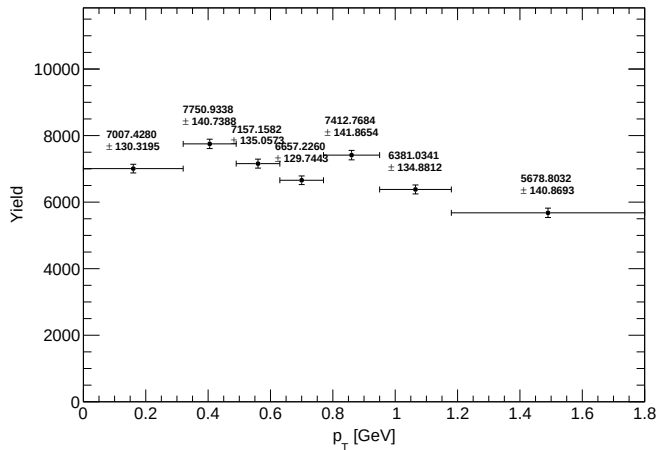
Corrected Yield (Total Error) (Flask)



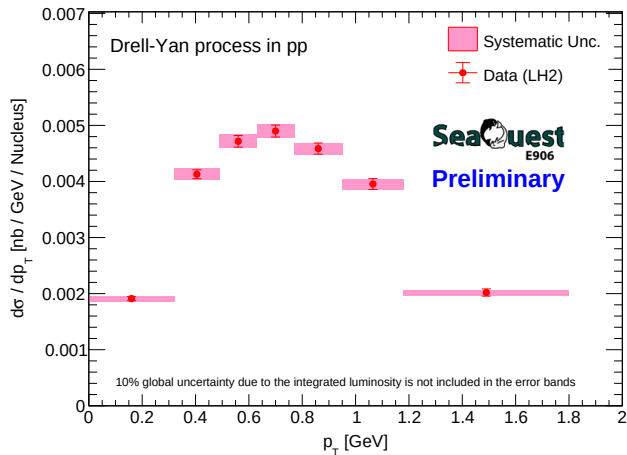
Corrected Yield (LH2 - Flask)



Corrected Yield (LD2 - LH2 - Flask)



CrossSection_LH2_vs_pT_with_logo



CrossSection_LD2_vs_pT_with_logo

